

Module 3 Lab 2: Textbook Problem 2.1

CS 82A - Introduction to Data Science

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GitHub: https://github.com/ramishatasfia5156-commits/cs82a-coursework/blob/main/module3/module3_lab2.ipynb

```
In [1]: import pandas as pd

df = pd.read_csv('usarrests.csv', index_col=0)
df.index.name = 'State'

print('Shape:', df.shape)
print()
print('Missing values per column:')
print(df.isna().sum())
```

Shape: (50, 3)

Missing values per column:

Murder 0

Assault 1

UrbanPop 0

dtype: int64

Summary Statistics

```
In [2]: df.describe().T[['min', '25%', '50%', 'mean', '75%', 'max']].round(2)
```

```
Out[2]:
```

	min	25%	50%	mean	75%	max
Murder	0.8	4.08	7.25	7.79	11.25	17.4
Assault	45.0	109.00	159.00	182.18	249.00	879.0
UrbanPop	6.0	53.25	66.00	74.20	77.75	570.0

Flagged Values

```
In [3]: valid_range = {'Murder': (0, 100_000), 'Assault': (0, 100_000), 'UrbanPop': (0, 100)}

flags = []
for col, (lo, hi) in valid_range.items():
    s = df[col].dropna()
    q1, q3 = s.quantile(0.25), s.quantile(0.75)
    iqr = q3 - q1
```

```

for state, val in s.items():
    out_of_range = val < lo or val > hi
    beyond_fence = val < q1 - 1.5 * iqr or val > q3 + 1.5 * iqr
    if out_of_range or beyond_fence:
        flags.append({'State': state, 'Column': col, 'Value': val,
                      'Outside valid range': out_of_range,
                      'Outside IQR fence': beyond_fence})

pd.DataFrame(flags)

```

```

Out[3]:

```

	State	Column	Value	Outside valid range	Outside IQR fence
0	South Carolina	Assault	879.0	False	True
1	Iowa	UrbanPop	570.0	True	True
2	New York	UrbanPop	6.0	False	True

```

In [4]: # Context for each flag
print(df.loc[['Iowa', 'Nebraska', 'Kansas', 'Minnesota']], '\n')
print(df.loc[['New York', 'Connecticut', 'New Jersey', 'Massachusetts',
              'Rhode Island']], '\n')

ratio = df['Assault'] / df['Murder']
print('Next-highest Assault:', df['Assault'].drop('South Carolina').max())
print('SC Murder rank:', int(df['Murder'].rank(ascending=False)['South
Carolina']))
print(f'SC Assault/Murder ratio: {ratio["South Carolina"]:.1f}    Median
state: {ratio.median():.1f}')

```

	Murder	Assault	UrbanPop
State			
Iowa	2.2	56.0	570
Nebraska	4.3	102.0	62
Kansas	6.0	115.0	66
Minnesota	2.7	72.0	66

	Murder	Assault	UrbanPop
State			
New York	11.1	254.0	6
Connecticut	3.3	110.0	77
New Jersey	7.4	159.0	89
Massachusetts	4.4	149.0	85
Rhode Island	3.4	174.0	87

```

Next-highest Assault: 337.0
SC Murder rank: 5
SC Assault/Murder ratio: 61.0    Median state: 22.9

```

Verdicts

Flag 1: Iowa, UrbanPop = 570 → IMPOSSIBLE.

The percentage of the state's population living in urban areas cannot be greater than 100%, therefore regardless of what else the numbers are for that particular

record, the UrbanPop value will fail the logical-range test. The 570 is a possible urban population percentage of 57% with a zero added to the end. Also, 57% sits right beside Iowa's Midwest neighbors, with Nebraska at 62%, Kansas at 66%, and Minnesota at 66%. **Action:** changed to 57; verified using the original published dataset.

Flag 2: New York, UrbanPop = 6 → NOT impossible, but improbable (extreme low outlier).

Six percent is a valid urban population percentage, thus it passes the range test. However, it fails the IQR fence. Context also rules it out since New York contains the country's largest city, while every other highly urban Northeastern state falls within a range of 77% to 89%. A single dropped digit from 86 explains the error. **Action:** corrected to 86; verified using the original source. Without a source, I would have converted it to NaN and then imputed rather than keeping a value I believe is incorrect.

Flag 3: South Carolina, Assault = 879 → NOT impossible, but extremely unusual and probably a data entry error.

With 879 arrests per 100,000, which is under 1% of residents, it is theoretically possible. Yet, 879 is 2.6x higher than North Carolina's Assault rate, which is the next-highest at 337. While South Carolina's murder rate of 14.4 ranks 5th and is high, it is still not exceptional. Therefore, South Carolina's Assault-to-murder ratio of 61 is about 2.7x the median state's ratio of 22.9. One changed digit, 879 instead of 279, explains the error. **Action:** corrected to 279; verified with the original source. Had the original source confirmed 879 as correct, I would have left it intact because a real extreme value is information, not noise.

Missing value: Georgia, Assault = NaN. Filled with 211, the value in the original published dataset (R's `datasets::USArrests`).

```
In [5]: df.loc['Iowa', 'UrbanPop'] = 57
df.loc['New York', 'UrbanPop'] = 86
df.loc['South Carolina', 'Assault'] = 279
df.loc['Georgia', 'Assault'] = 211

print('Missing values:', df.isna().sum().sum())
df.describe().T[['min', '25%', '50%', 'mean', '75%', 'max']].round(2)
```

Missing values: 0

```
Out[5]:
```

	min	25%	50%	mean	75%	max
Murder	0.8	4.08	7.25	7.79	11.25	17.4
Assault	45.0	109.00	159.00	170.76	249.00	337.0
UrbanPop	32.0	54.50	66.00	65.54	77.75	91.0

Urban Population Categories

```
In [6]: labels = ['small (<50%)', 'medium (50-59%)', 'large (60-69%)', 'extra-large (70%+)']
df['UrbanCategory'] = pd.cut(df['UrbanPop'], bins=[0, 50, 60, 70, 101],
                              labels=labels, right=False)

df.groupby('UrbanCategory', observed=True).agg(
    states=('UrbanPop', 'size'),
    avg_murder=('Murder', 'mean'),
    avg_assault=('Assault', 'mean')).round(1)
```

Out[6]:

	states	avg_murder	avg_assault
UrbanCategory			
small (<50%)	8	8.2	174.8
medium (50-59%)	9	6.7	127.6
large (60-69%)	12	7.8	153.5
extra-large (70%+)	21	8.1	197.6

Cleaning Log

Change	Before	After	Reason
Iowa UrbanPop	570	57	Impossible: percent cannot exceed 100
New York UrbanPop	6	86	Improbable low outlier; verified against source
South Carolina Assault	879	279	Extreme outlier, likely entry error; verified against source
Georgia Assault	NaN	211	Missing; filled from source
Murder	—	—	Screened; no flags
UrbanCategory	—	new column	Textbook part c

AI Disclosure

AI Disclosure: Claude (Anthropic) was used to help explain concepts and draft initial code for this lab. All code was reviewed, retyped, and understood by me. I can explain how `describe()` and the IQR fence rule flag suspicious values, why Iowa's 570 is impossible while New York's 6 and South Carolina's 879 are only extreme, and how `pd.cut` groups states into urban population categories.